PREFACE

This project represents the culmination of extensive research, experimentation, and collaboration with a focus on land cover detection and semantic segmentation. The objective is to develop and implement advanced methodologies that allows the enhancement of accuracy, efficiency, with usability of geospatial analytical tools. By integrating computer vision machine learning models and sophisticated data augmentation techniques from Albumentations, we have created a robust framework for land cover detection. Containerization with Docker ensures a consistent and reproducible environment, simplifying deployment and scaling. Furthermore, the deployment of a user-friendly Streamlit dashboard allows for real-time visualization and interaction with model predictions, making the technology accessible to a wider audience. This project also underscores the importance of model reliability and adaptability over time. By incorporating robust drift detection mechanisms, such as Kullback-Leibler (KL) aka KL divergence and cross-entropy methods, we ensure continuous monitoring and adjustment of the system, maintaining predictive accuracy as data evolves. The work presented here is a testament to the potential of interdisciplinary collaboration and the transformative impact of machine learning in environmental monitoring and urban planning. It provides a robust framework that can be adapted for various applications within the geospatial domain. The journey of this project has been both challenging and rewarding, made possible through the unwavering support of several individuals. I am profoundly grateful to my professor, Mr. Sahil K. Shah, for his invaluable mentorship and guidance. I do express sincere thanks to my family, to my friends for being a source of continuous encouragement with support. Through this preface, I aim to convey the dedication, effort, and collaborative spirit invested in this project. It is my sincere hope that the findings and methodologies presented here will contribute meaningfully to the advancement of machine learning applications in geospatial analysis and inspire further innovation in this exciting field.